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Glycopyrrolate: It's time to review☆

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Abstract Medication shortages have become an all-too-common inconvenience that has forced anesthesia providers to examine our administering practices. Because of these shortages, commonly used medications are at the greatest risk. Glycopyrrolate (Robinul), which has been in short supply in recent years, is one of the most widely used anticholinergic agents, especially in conjunction with the anticholinesterase neostigmine, for reversal of neuromuscular blockade (NMB) drugs. Here we review multiple studies from 1972 through 1986 that used varying methods of patient selection and dosage and drug combination criteria, and which noted that glycopyrrolate had a superior efficacy and adverse effect profile when compared with atropine in NMB reversal. Meta-analysis from these studies indicated that the dosage of 0.2 mg of glycopyrrolate for every 1 mg of neostigmine, given concomitantly (maximum 1 mg glycopyrrolate and 5 mg neostigmine), demonstrated the greatest efficacy with the lowest incidence of unwanted adverse effects. It has now become common practice to use a dosage ratio of 0.2 mg glycopyrrolate to 1.0 mg neostigmine for NMB reversal. Yet since 1986, there have been no studies on reversal with glycopyrrolate and neostigmine. Frequent medication shortages and good medical practice should be an impetus for clinicians to reevaluate dosing practices of critical medications and revisit these drugs, such as glycopyrrolate, with more current studies.

Medication shortages have become an all too common inconvenience that has forced anesthesia providers to examine our administering practices. The most routinely used medications are at the greatest risk. Glycopyrrolate (Robinol) has been in short supply in recent years and is one of the most widely used anticholinergic agents, particularly in conjunction with neostigmine, an anticholinesterase used for reversal of neuromuscular blockade drugs (NMBD). Multiple studies from 1972 through 1986, using varying methods of patient selection, dosage and drug combinations, concluded that glycopyrrolate had a superior efficacy and side effect profile when compared to atropine in NMBD reversal. Meta-analysis from these studies indicated that the dosage of 0.2 mg of glycopyrrolate for every 1 mg of neostigmine, given concomitantly, (maximum 1 mg glycopyrrolate and 5 mg Neostigmine) demonstrated the greatest efficacy with the lowest incidence of unwanted side effects. It has now become common practice to use a dosage ratio of 0.2 mg glycopyrrolate to 1.0 mg neostigmine for NMB reversal. However, since 1986 there have been no studies on reversal with glycopyrrolate and neostigmine. Frequent medication shortages and good medical practice should be an impetus for clinicians to re-evaluate dosing practices of critical medications and revisit these drugs, such as glycopyrrolate, with more current studies.

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In 1960, Frank and Lunsford studied glycopyrrolate and discovered it to be a potent, long acting anti-cholinergic agent with many potential clinical applications [1]. In 1969, during his investigation into sudden cardiac arrest during adenotonsillectomies, Watson used glycopyrrolate in the perioperative period to raise the pH of gastric secretions [2]. He found that glycopyrrolate had a high efficacy for decreasing the volume and acidity of gastric secretions and that unexpected aspirations during and after had almost no sequelae [3]. At the conclusion of the study, Watson was impressed with glycopyrrolate's efficacy and minimal side effect profile and proposed it be used as a replacement for atropine for reversal of NMBD with neostigmine at half the dose of atropine [2]. In 1972, Klingenmaier studied the effects of various mixtures of glycopyrrolate and neostigmine, comparing it to atropine and neostigmine for NMBD reversal on 49 patients [3]. The study concluded that 0.2 mg of glycopyrrolate mixed with 1 mg of neostigmine, when given in combination, demonstrated a significantly lower incidence of arrhythmias and changes in heart rate when compared to the atropine group [3]. That same year Ramamurthy studied 85 adult patients to evaluate whether glycopyrrolate had significant advantages over atropine. It was concluded that glycopyrrolate provided equal protection against neostigmine induced bradycardia as atropine, while having significantly fewer incidences of tachycardia and other arrhythmias associated with atropine usage [4]. Ramamurthy found that a dose of 0.4 mg to 2 mg of neostigmine provided the best compromise between unwanted tachycardias and prevention of neostigmine induced bradycardia [4]. However, the conclusion was based on only five patients in each of the groups [4].

A 1977 study by Mirakhur, Dundee and Clarke compared 2 different dosage ratios of neostigmine and glycopyrrolate and compared it to equipotent dosages of neostigmine and atropine [5]. The heart rates and antisialogogue effects were superior in the glycopyrrolate group and a dosage of 2.5 mg neostigmine and 1.2 mg glycopyrrolate had the most stable heart rate and fewest incidences of arrhythmia [5]. The authors noted that this confirmed the conclusion reached by previous studies by Ramamurthy [4,5]. A similar study in 1977 by Ostheimer reviewed the established 0.2 mg to 1 mg ratio in 2 groups, one being dosed simultaneously and the other with glycopyrrolate 5 minutes after neostigmine was given. These results were compared to a group given 1 mg atropine and 2.5 mg of neostigmine. As concluded on previous studies the effectiveness of reversal with fewer cardiac side effects was demonstrated within the glycopyrrolate group [6]. Within those 2 groups, the simultaneous dosage was superior [6]. This type of dosing was studied again by Mirakhur and Dundee in a 1981 study of atropine and glycopyrrolate given before or in a mixture with neostigmine [7]. It concluded again that glycopyrrolate (10 μg/kg), when administered simultaneously in a mixture with neostigmine was associated with the most stable heart rates [7]. The authors noted that from the data obtained from this study agreed with the previous Ramamurthy findings that a dose of 0.2 mg glycopyrrolate per mg of neostigmine seemed to be justified [4,7].

A 1980 study by Bali and Mirakhur comparing glycopyrrolate-neostigmine and atropine-neostigmine mixtures for NDMB reversal in closed mitral valvotomy demonstrated both to be effective reversal agents but glycopyrrolate had fewer initial increases in heart rate [8]. Another study in 1980, by Cozanitis, Dundee, Merrett, Jones and Mirakhur was a 28 center, double blind study with 641 patients over a 6-month period. Patients were given weight-based doses of either 0.05 mg/kg neostigmine, 0.02 mg/kg atropine or 0.01 mg/kg glycopyrrolate [9]. This study largely agreed with previous studies and found that while both were effective reversal agents, the glycopyrrolate group demonstrated smaller changes in heart rate than those who had received atropine [9]. This result was particularly more noticeable in the patients with cardiovascular disease [9]. It differed from Mirakhur, Dundee and Clarke (1977) in that they failed to show the initial increase in heart rate that was demonstrated in this study and it was surmised that the difference was due to larger mean doses secondary to the weight based doses [5,9]. As with all previous studies, it concluded that glycopyrrolate proved to be a superior to atropine [9]. The 5 μg/kg dose of glycopyrrolate did not adequately prevent unwanted bradycardias and there was little benefit found from increasing the dose to 15 μg/kg[9]. Therefore, the 10 μg/kg, which largely justified the ratio of 0.2 mg per 1 mg of neostigmine first noted by Ramamurthy in 1972, was the optimum dose demonstrating the most stable heart rates [4,9]. A separate 1982 study of 84 patients by Kongrud and Sponheim showed that 0.5 mg of glycopyrrolate with 2.5 mg of neostigmine had a superior side effect profile while protecting against neostigmine induced bradycardia and arrhythmias when compared to 1 mg to 2.5 mg atropine and neostigmine [10].

In 1985 Salem, et al. studied 40 patients undergoing open-heart surgery for valve replacement or coronary artery bypass grafting [11]. The authors had taken note of the Conzanitis study, which had concluded that cardiovascular patients had a marked difference response to atropine and glycopyrrolate versus non-cardiac patients [9,11]. They also took into account the Bali and Mirakhur conclusion in 1980 that there was a difference for glycopyrrolate versus atropine in open mitral valvotomy patients [8,11]. Patients that had not received β-blocker therapy demonstrated fewer initial tachycardias and better protection of muscarinic effects of neostigmine versus the group receiving atropine [11]. However, when β-blockers were given concomitantly there was no significant difference between the groups [11].

The most recent study to date on the subject was the 1986 Salem and Ahearn study of 150 patients separated in 4 groups whom all received 5 mg of neostigmine and then either 1.2 or 1.8 mg of atropine versus 0.6 or 0.9 mg of glycopyrrolate [12]. The authors noted that the impetus for the study was what they determined to be a lack of data on the optimum adult dose of glycopyrrolate or atropine to be used with 5 mg of neostigmine [12]. The 0.9 mg of glycopyrrolate showed the most stable heart rate from baseline with the fewest side effects [12]. It was also noted that because of the duration of action of glycopyrrolate, it is superior when using larger doses of neostigmine compared to atropine [12].

Salem and Ahearn's 1986 study was the last to specifically investigate varying dosages of glycopyrrolate for reversal with neostigmine. More recently there have been several studies of note that have loosely broached this topic. A 1991 study by Wetterslev [13] looked at 55 patients that were given gallamine reversal, used singular reversal doses of either atropine or glycopyrrolate and monitored patients for 1 hour post operatively for any heart rate or ECG changes. Only four patients in the atropine group had incidences of cardiac arrhythmias and thus suggested that this combination be avoided in cardiac patients. In 1992, a study by Salmenpera [14] compared postoperative nausea and vomiting incidences between singular reversal doses glycopyrrolate and atropine, with glycopyrrolate having a higher incidence. A 1997 study by Vlymenn [15] used singular reversal dosages of either atropine-neostigmine or glycopyrrolate-neostigmine. The 2 groups were then monitored for parasympathetic nervous system variability. The atropine group demonstrated persistent baroreflex sensitivity impairment and high frequency of heart rate variability. Most recently, a 2014 study in Thailand by Ittichaikulthol [16] compared heart rate variabilities with atropine and glycopyrrolate. Atropine is still the most widely used anticholinesterase in Thailand, but glycopyrrolate is now becoming more available. In this study, 51 patients given either a single 2.5 mg/1.2 mg neostigmine/atropine reversal or 2.5 mg/0.6 mg/0.2 mg neostigmine/atropine/glycopyrrolate. Heart rate variabilities from baseline were compared between the 2 groups and no significant difference was found. While these studies investigate side effects and singular reversal dosages of glycopyrrolate, they do not directly investigate alternative reversal dosages of glycopyrrolate and neostigmine.

As practitioners, we are forced to re-evaluate medication dosages, medication type/class and/or alternatives. Considering the age and small sample sizes of these studies, do we still need to keep our archaic practices or is there room to update them? This most common medication, glycopyrrolate, represents a need to probe further.

References

[1] Frank BV, Lunsford CD. Derivatives of 3-pyrrolidinols. III: the chemistry, pharmacology and toxicology of some N-subsituted-3-pyrrolidyl α-substituted phenylacfetates. J Med Pharm Chem 1960;2:523.

[2] Boatright CF, Newell RC, Watson RL, Barnhart RA. Sudden cardiac arrest in Adenotonsillectomy. Tr Am Acad Ophth Otol 1970;74:1139-45.

[3] Klingenmaier C, Herman RB, Thompson D, Watson R. Reversal of neuromuscular blockade with a mixture of neostigmine and glycopyrrolate. Anesth Analg 1972;51.3:468-72.

[4] Ramamurthy S, Shaker MH, Winnie AP. Glycopyrrolate as a substitute for atropine in neostigmine reversal of muscle relaxant drugs. Can Anaesth Soc J 1972;19.4:399-410.

[5] Mirakhur RK, Dundee JW, Clarke RSJ. Glycopyrrolate—neostigmine mixture for antagonism of neuromuscular block: comparison with atropine—neostigmine mixture. Br J Anaesth 1977;49.8:825-9.

[6] Ostheimer GW. A comparison of glycopyrrolate and atropine during reversal of Nondepolarizing neuromuscular block with neostigmine. Anesth Analg 1977;56.2:182-6.

[7] Mirakhur RK, Dundee JW, Jones CJ, Coppel DL, Clarke RSJ. Reversal of neuromuscular blockade. Anesth Analg 1981;60.8:557-62.

[8] Bali IM, Mirakhur RK. Comparison of glycopyrrolate, atropine and Hyoscine in mixture with neostigmine for reversal of neuromuscular block following closed mitral valvotomy. Acta Anaesthesiol Scand 1980;24.4:331-5.

[9] Cozanitis DA, Dundee JW, Merrett JD, Jones CJ, Mirakhur RK. Evaluation of glycopyrrolate and atropine as adjuncts to reversal of non-depolarizing neuromuscular blocking agents in a “true-to-life” situation. Br J Anaesth 1980;52.1:85-9.

[10] Kongsrud F, Sponheim S. A comparison of atropine and glycopyrrolate in Anaesthetic practice. Acta Anaesthesiol Scand 1982;26.6:620-5.

[11] Salem MG, Richardson JC, Meadows GA, Lampluch G, Lai KM. Comparison between glycopyrrolate and atropine in a mixture with neostigmine for reversal of neuromuscular blockade. Br J Anaesth 1985;57.2:184-7.

[12] Salem MG, Ahearn RS. Atropine or glycopyrrolate with neostigmine 5 Mg: a comparative dose–response study. J R Soc Med 1986;79:19-21.

[13] Wetterslev J, Jarnvig I, Jøosrgensen LN, Olsen NV. Split-dose atropine versus glycopyrrolate with neostigmine for reversal of gallamine-induced neuromuscular blockade. Acta Anaesthesiol Scand 1991;35.5:398-401.

[14] Salmenperä M, Kuoppamäki R, Salmenperaä A. Do anticholinergic agents affect the occurrence of Postanaesthetic nausea? Acta Anaesthesiol Scand 1992;36.5:445-8.

[15] Van Vlymen JM, Parlow JL. The effects of reversal of neuromuscular blockade on autonomic control in the perioperative period. Anesth Analg 1997;84.1:148-54.

[16] Ittichaikulthol W, Pisitsak C, Wirachpisit N, Piathong P, Suyawet R, Komonhirun R. A comparison of the combination of atropine and glycopyrrolate with atropine alone for the reversal of muscle relaxant. J Med Assoc Thai 2014;97.7:705-9.